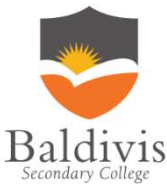
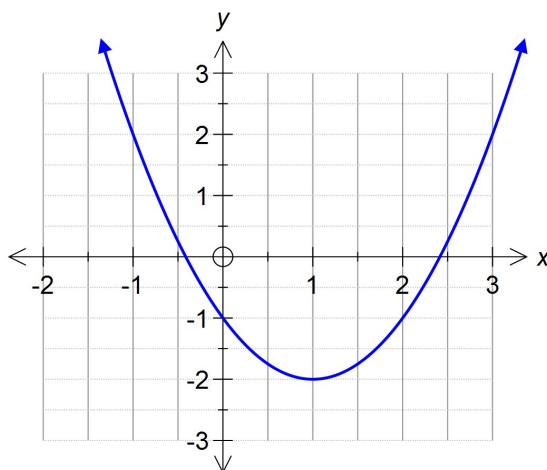


Name:			Date:	
Teacher:				
	Year 9 Mathematics Mini test 5 Topic – Quadratic Equations		SCORE: / 25	
	<u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	30 minutes			
Weighting:	5 %			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.			

Multiple Choice Section

5 marks

Questions 1 – 3 refer to the graph below:



1 The equation of the axis of symmetry for this graph is:

A $x = 1$

B $x = -2$

C $y = 1$

D $y = -2$

E $y = -1$

2 The turning point of the graph is:

A a minimum at $(-2, 1)$

B a maximum at $(-2, 1)$

C a minimum at $(1, -2)$

D a maximum at $(1, -2)$

E a minimum at $(-1, 2)$

3 The equation of the graph is:

A $y = (x + 1)^2 - 2$

B $y = (x - 1)^2 + 2$

C $y = (x - 2)^2 + 1$

D $y = (x - 1)^2 - 2$

E $y = (x + 1)^2 + 2$

- 4 The transformation of the graph $y = x^2$ to $y = x^2 + 2$ is described by:
- A a translation of 2 units left **B a translation of 2 units up**
- C a translation of 2 units right D a translation of 2 units down
- E a translation of 2 units right and 2 units down

- 5 Compared to $y = x^2$, the narrowest graph is:

- A $y = 2.5x^2$ B $y = \frac{1}{5}x^2$ C $y = 5x^2$
- D $y = 25x^2$** E $y = 0.5x^2$

Short Answer Section

15 marks

1

[11 marks: 7, 4]

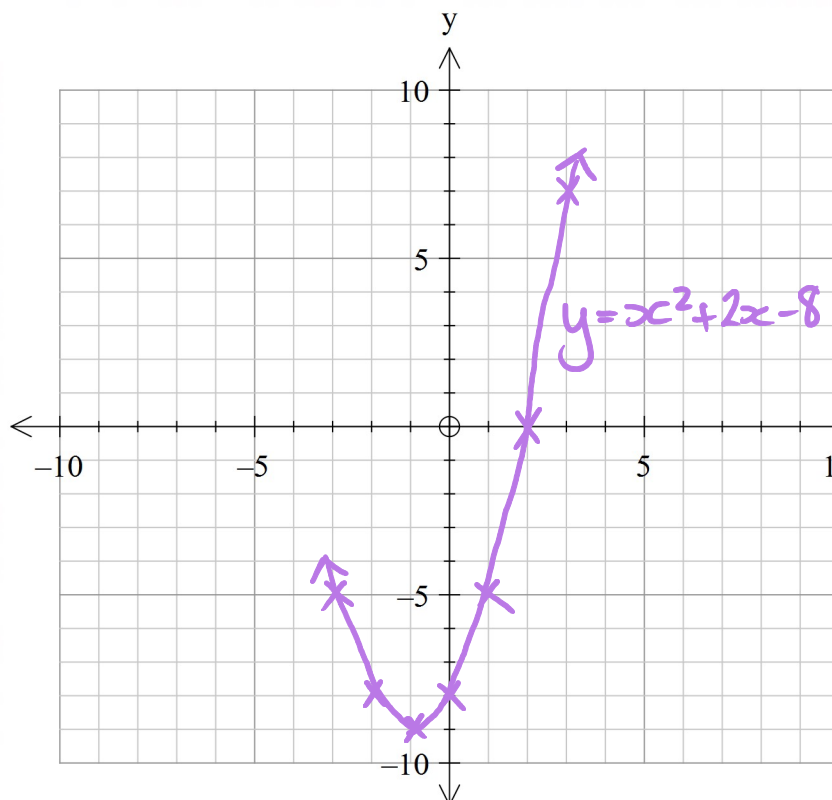
For the quadratic equation $y = x^2 + 2x - 8$.

- a Create a table of values with x-values from -3 to 3

x	-3	-2	-1	0	1	2	3
y	-5	-8	-9	-8	-5	0	7

1 mark for each correct value of y calculated

- b Using your table of values, construct the graph of the parabola including all major features.



1 mark correct co-ordinates

1 mark smooth parabolic curve

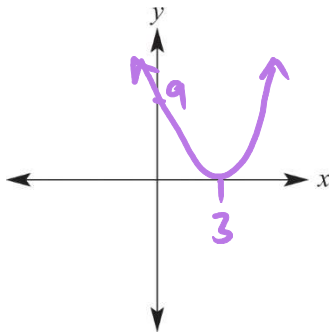
1 mark arrows

1 mark label.

Sketch the graphs of these rules showing the coordinates of the turning point and the y -intercept.

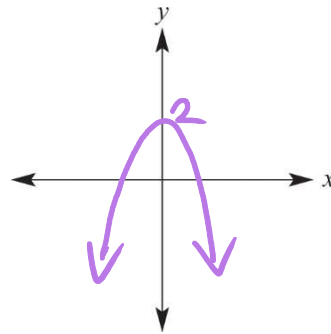
a $y = (x - 3)^2$

T.P. = $(3, 0)$ $\frac{1}{2}$ mark
 y -int = $(0, 9)$ $\frac{1}{2}$ mark



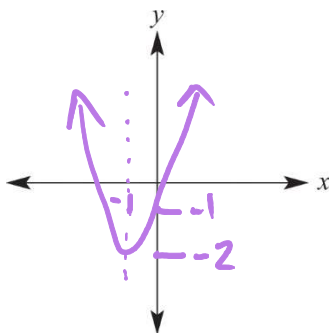
b $y = -x^2 + 2$

T.P. = $(0, 2)$ $\frac{1}{2}$ mark
 y int = $(0, 2)$ $\frac{1}{2}$ mark



c $y = (x + 1)^2 - 2$

T.P. = $(-1, -2)$ 1 mark
 y int = $(0, -1)$ 1 mark



3

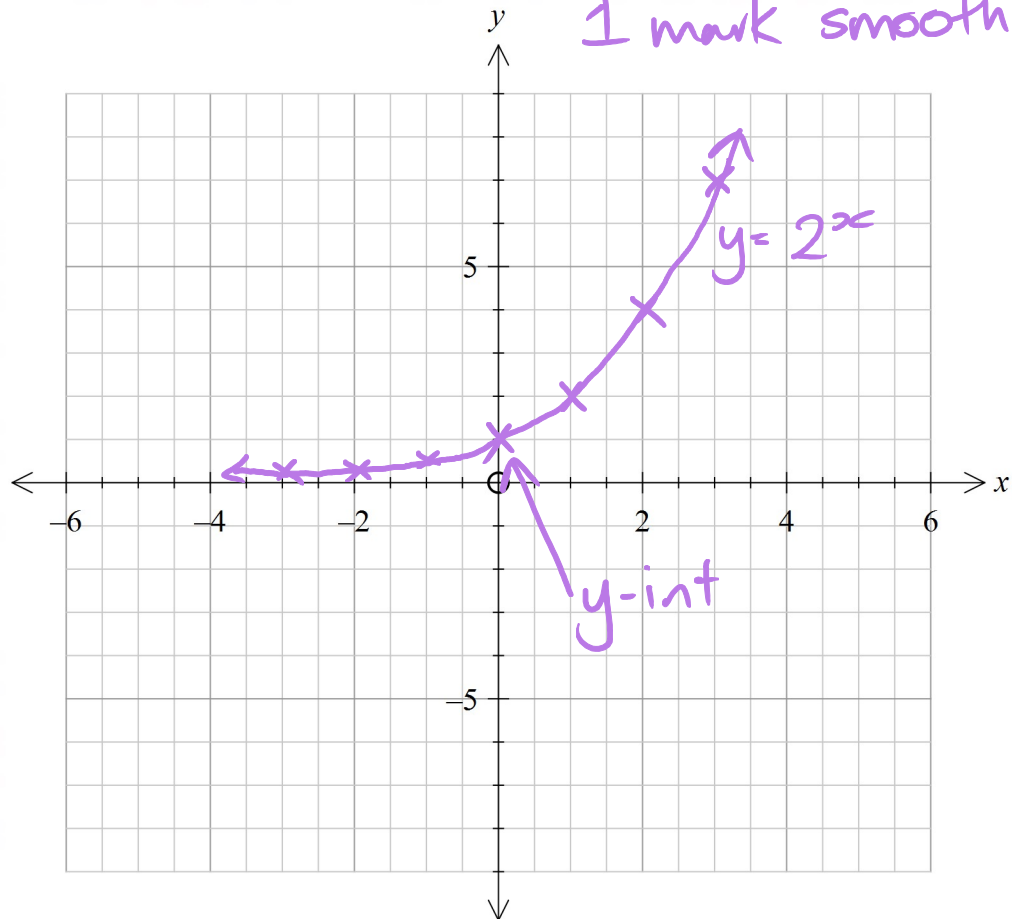
[5 marks: 2, 2, 1]

Sketch the graphs of these rules showing the coordinates of the turning point and the y -intercept.

a Complete the following table of values for $y = 2^x$.

x	-3	-2	-1	0	1	2	3
y	0.125	0.25	0.5	1	2	4	8

b Draw the graph of $y = 2^x$.



c Clearly label the y -intercept of $y = 2^x$.

1 mark for correct labelling